

Dropline Four Connect - Evaluation Report

Golden Oracle, model outcomes, criterion-level results and evidence limitations

Harbor task: turing/dropline-four-connect | version 1.0.0 | prepared 14 September 2026

Recorded platform outcomes on the frozen r6 package. No score adjustment or fresh paid run was performed during delivery preparation. Interpretation limits are stated below.

Headline

Oracle scored 0.9917 overall: all 42 Functional criteria passed, with Render, Constraints and Polish also at 1.0. Visual was 0.9583. Only visual_spacing_and_layout received 4/5 rather than 5/5: the judge noted imbalance in the wide completed-matches card and tall analysis workspace. This is not an overall 1.0 result. The prior analysis pending-input and restart observations passed in this run.

Run	Overall	Functional	Polish	Visual	Recorded status
Oracle	0.9917	1.0000	1.0000	0.9583	42/42; both gates pass
GPT-5.4 mini (high)	0.1766	0.0305	0.0000	0.7917	1/42; both gates pass
Gemini 3.7 Flash	0.7804	0.8550	0.5455	0.7917	35/42; both gates pass
Claude Haiku 4.5	0.0000	0.0000	0.0000	0.5833	0/42; Constraints = 0
No-op	0.0000	0.0000	0.0000	0.0000	graded=0; no_op=1

Render is 1.0 for all four built applications. Constraints is 1.0 for Oracle, GPT-mini and Gemini, and 0.0 for Haiku. No-op zeros are control outputs, not 58 independently observed failures.

Task description and verifier design

Area	Coverage
Identity and durable games	Two seeded accounts; protected bearer sessions and account-wide logout; exact boards, scores, histories, revisions and private archives.
Connect Four and replay	Gravity, full columns, wins in every direction and both colors, exact draw, terminal lock, reversible scores, redo branching, latest-ten archive and read-only replay.
Private branching studies	Fork any replay position; frozen source snapshot; nested alternatives, child reuse, rename, selection, comparison and competitive-score isolation.
Transactional subtree transplant	Preview every derived position, recursive identity-preserving merges, same-study frozen traversal, illegal-descendant rollback, dual-revision conflicts and durable retry receipts.
Bounded tactical proofs	Depths 1-4, terminal-before-horizon evaluation, adversarial win/loss/draw/unknown outcomes, shortest wins, longest resistance, complete ordered proof and browsable board positions.
Persistence and presentation	Two process restarts, SQLite state and receipt durability, input rejection/ownership; keyboard/focus, pending feedback, reduced motion and 375px layouts; six Visual axes.

Runtime: Node.js/Express on 0.0.0.0:3000, vanilla browser JavaScript and SQLite at /app/dropline.db; entry node /app/server.js. Public agent and separate-verifier networks. Provider credentials are supplied by the platform, never embedded in the task. Codex / gpt-5.6-luna / max reasoning, temperature 0.

Dimension	Criteria	Budget	Aggregation / role
Render	1	600 s	all_pass gate
Constraints	2	600 s	all_pass gate
Functional	42	9000 s	weighted mean / 60%
Polish	7	900 s	weighted mean / 20%
Visual	6	900 s	weighted mean / 20%

If Render <= 0 or Constraints <= 0: Overall = 0. Otherwise: Overall = 0.6 x Functional + 0.2 x Polish + 0.2 x Visual. Scores are rounded to four decimals. Functional counts are not weighted scores. Visual raw 4/5 normalizes to 0.75 and raw 5/5 to 1.0; the gate handles an unreviewable page separately.

Observed grading time

Run	All-verifier elapsed	Exported status
Oracle	49m 51s	No trial-level exception
GPT-5.4 mini (high)	20m 52s	No trial-level exception
Gemini 3.7 Flash	54m 42s	No trial-level exception
Claude Haiku 4.5	20m 07s	No trial-level exception

These are complete verifier-phase elapsed times, not per-criterion budgets. Oracle completed in about 50 minutes. No trial-level timeout was recorded, but GPT has an individual tool timeout in its sign-out explanation. Platform static/rubric QC output was not supplied with these trial exports; successful execution is not a substitute for claiming 53/53 QC evidence.

Findings and limitations

Oracle

Oracle scored 0.9917 overall: all 42 Functional criteria passed, with Render, Constraints and Polish also at 1.0. Visual was 0.9583. Only visual_spacing_and_layout received 4/5 rather than 5/5: the judge noted imbalance in the wide completed-matches card and tall analysis workspace. This is not an overall 1.0 result. The prior analysis pending-input and restart observations passed in this run.

GPT-mini

GPT-mini scored 0.1766 with both gates passed. Only the weight-2 seeded_state_import_and_isolation criterion passed Functional (2/65.5 = 0.0305). The exported UI and manifest omit New game, a required workflow entry point; many later failures are blocked setup, not separately demonstrated algorithm defects. Polish also reports absent replay Previous/Next/Close controls and a TypeError leaving analysis boards and selectors empty.

Gemini

Gemini scored 0.7804 and passed 35/42 Functional criteria. Recorded failures concern keyboard Undo/Redo focus, malformed operation identifiers in analysis and transplant commits, tactical defence/loss calculations, full-column recommendations, stale report invalidation and extra report fields. Polish deductions concern clipped mobile analysis, repeated pending commit activation and lost focus/feedback. Its overall score remains above 0.7; only GPT-mini meets the stated below-0.7 target in this comparison.

Haiku

Haiku scored 0.0000 because `local_refresh_and_health` failed: the judge observed HTTP 200 health/game responses but a blank authenticated page and a null-status `TypeError` after reload. This is a Constraints-gated zero, not absence of all working code. Its Functional judge recorded inverted row coordinates, stale winner display, draw-archive Undo errors and missing analysis/transplant/tactical surfaces. Some multi-tab checks used shared storage rather than the requested isolated contexts and were unverified.

- GPT analysis creation uses a native browser prompt for the name. The Functional explanation that the creation name field was hidden does not match that code path; the hidden field is for renaming. A dismissed/unhandled native prompt can produce the reported name-required message. Without exported detailed judge actions, that creation failure is not a cleanly established missing-input defect. Other dimensions report a separate analysis-render `TypeError`; these are distinct observations.
- GPT `cross_tab_sign_out_revocation` observed distinct tokens, proactive sign-out and old-token 401 rejection, but post-revocation re-login verification was unfinished after a tool timeout. Do not describe the failed criterion as proof that logout revocation is broken.
- Missing New game and inaccessible studies cause many dependent GPT failures. Their number cannot be interpreted as independent evidence of 41 different algorithm defects. The recorded score is preserved, not corrected or presented as a complete isolated test of its backend.
- Haiku multi-tab evidence was limited by a shared-context reload failure. Its main reload defect is also reported separately by Constraints; the unexecuted distinct-session probes are not additional demonstrated session bugs.
- The exports contain all criterion verdicts and no trial-level exceptions, but that does not mean every blocked or unverified subcheck executed. This delivery is not blanket fairness certification. No local browser regrade of these model submissions was performed during final packaging.
- One current exported attempt per model is included. This is not a repeatability study. Oracle is above 0.95 with Functional 1.0, not overall 1.0; Gemini is above 0.7 and Haiku is a gate-zero. Administrative acceptance is not asserted.

All 58 verifier outcomes

These reproduce exported verdicts without changing scores. Fail includes blocked/unverified observations where the judge assigned zero; it does not always mean an independently exercised defect. Raw Visual values are shown as /5.

Render

Criterion	Weight	Oracle	GPT-mini	Gemini	Haiku
page load and refresh	1.0	Pass	Pass	Pass	Pass

Constraints

Criterion	Weight	Oracle	GPT-mini	Gemini	Haiku
required server backed workspace	1.0	Pass	Pass	Pass	Pass
local refresh and health	1.0	Pass	Pass	Pass	Fail

Functional

Criterion	Weight	Oracle	GPT-mini	Gemini	Haiku
seeded state import and isolation	2.0	Pass	Pass	Pass	Fail
completed match archive and replay	3.0	Pass	Fail	Pass	Fail
gravity turns and full column	1.0	Pass	Fail	Pass	Fail
horizontal and vertical wins	1.0	Pass	Fail	Pass	Fail
both diagonal wins	1.0	Pass	Fail	Pass	Fail
yellow win and terminal lock	1.0	Pass	Fail	Pass	Fail
exact draw and terminal lock	1.0	Pass	Fail	Pass	Fail
move history undo redo branching	2.0	Pass	Fail	Pass	Fail
new game reset and persistence	1.0	Pass	Fail	Pass	Fail
keyboard focus and activation	2.0	Pass	Fail	Fail	Fail
multi tab revision and duplicate guard	2.0	Pass	Fail	Pass	Fail
cross tab sign out revocation	1.0	Pass	Fail	Pass	Fail
accessible grid and winning names	1.0	Pass	Fail	Pass	Fail
analysis fork exact prefix	2.0	Pass	Fail	Pass	Fail
analysis sibling paths and navigation	2.0	Pass	Fail	Pass	Fail
analysis nested continuations	1.5	Pass	Fail	Pass	Fail
analysis terminal score isolation	2.0	Pass	Fail	Pass	Fail
analysis draw and root boundary	1.5	Pass	Fail	Pass	Fail

Criterion	Weight	Oracle	GPT-mini	Gemini	Haiku
analysis position comparison	2.0	Pass	Fail	Pass	Fail
analysis ownership boundaries	2.0	Pass	Fail	Pass	Fail
analysis stale edit and recovery	2.0	Pass	Fail	Pass	Fail
analysis operation receipts	2.0	Pass	Fail	Pass	Fail
analysis rejections are atomic	1.0	Pass	Fail	Fail	Fail
analysis source snapshot independence	1.0	Pass	Fail	Pass	Fail
restart persistence and seed idempotence	1.0	Pass	Fail	Pass	Fail
runtime manifest routes	0.5	Pass	Fail	Pass	Fail
transplant preview rederived positions	2.0	Pass	Fail	Pass	Fail
transplant atomic commit mapping	2.0	Pass	Fail	Pass	Fail
transplant recursive merge identity	2.0	Pass	Fail	Pass	Fail
transplant same study frozen subtree	2.0	Pass	Fail	Pass	Fail
transplant illegal descendant rollback	2.0	Pass	Fail	Pass	Fail
transplant dual revision conflicts	2.0	Pass	Fail	Pass	Fail
transplant durable receipts	1.5	Pass	Fail	Pass	Fail
transplant private validation	1.0	Pass	Fail	Fail	Fail
tactics immediate and horizon	1.5	Pass	Fail	Pass	Fail
tactics forced fork search	2.0	Pass	Fail	Pass	Fail
tactics opponent defence and loss	2.0	Pass	Fail	Fail	Fail
tactics full columns and terminal draw	1.5	Pass	Fail	Fail	Fail
tactics complete bounded proof	2.0	Pass	Fail	Pass	Fail
tactics readonly repeatability	1.0	Pass	Fail	Fail	Fail
tactics server input boundaries	1.0	Pass	Fail	Fail	Fail
independent same name analyses	0.5	Pass	Fail	Pass	Fail

Polish

Criterion	Weight	Oracle	GPT-mini	Gemini	Haiku
archive replay usability	1.0	Pass	Fail	Pass	Fail
responsive layout and feedback	0.5	Pass	Fail	Fail	Fail
reduced motion usability	0.5	Pass	Fail	Pass	Fail

Criterion	Weight	Oracle	GPT-mini	Gemini	Haiku
control labels and focus	1.0	Pass	Fail	Pass	Fail
transplant control feedback	1.0	Pass	Fail	Fail	Fail
tactical proof keyboard inspection	0.5	Pass	Fail	Pass	Fail
analysis keyboard actions and feedback	1.0	Pass	Fail	Fail	Fail

Visual

Criterion	Weight	Oracle	GPT-mini	Gemini	Haiku
visual typography	1.0	5/5	4/5	5/5	4/5
visual color and contrast	1.0	5/5	5/5	5/5	4/5
visual spacing and layout	1.0	4/5	4/5	4/5	3/5
visual hierarchy and scanability	1.0	5/5	5/5	4/5	3/5
visual overall craft	1.0	5/5	3/5	4/5	3/5
visual responsive consistency	1.0	5/5	4/5	3/5	3/5

Exported non-full-credit explanations

The following are the judges' recorded explanations, not endorsed causal conclusions. In particular the GPT native-prompt and unfinished revocation observations are qualified in the findings above.

Oracle

visual / visual spacing and layout

Strong grouping and consistent gutters on game, replay, comparison, transplant, and tactical panels. Minor imbalance remains in the wide completed-matches card and the very tall analysis workspace.

GPT-5.4 mini (high)

functional / completed match archive and replay

No visible New game control exists, so the required empty Avery match and shared Red-win journey could not be established.

functional / gravity turns and full column

Blocked by missing visible New game control.

functional / horizontal and vertical wins

Blocked by missing visible New game control and unavailable shared Red-win setup.

functional / both diagonal wins

Blocked by missing visible New game control.

functional / yellow win and terminal lock

Blocked by missing visible New game control.

functional / exact draw and terminal lock

Blocked by missing visible New game control.

functional / move history undo redo branching

Seeded Undo/Redo worked, but the required New game and shared terminal legs were unavailable.

functional / new game reset and persistence

No visible New game button or New game request was observed.

functional / keyboard focus and activation

Blocked by missing required New game setup.

functional / multi tab revision and duplicate guard

Independent Avery contexts and distinct tokens were observed, but required New game/move/stale-revision probes were blocked by the missing New game control.

functional / cross tab sign out revocation

Distinct Avery tokens, proactive other-tab sign-out, and old-token 401 rejection were observed. The required post-revocation re-login verification was not completed after a tool timeout.

functional / accessible grid and winning names

Seeded accessible cell names were observed, but required empty active board and shared Red-win checkpoints were unavailable.

functional / analysis fork exact prefix

Avery Red source could not be created because New game was unavailable.

functional / analysis sibling paths and navigation

Required Avery Red-source study was unavailable.

functional / analysis nested continuations

Required shared analysis setup was unavailable.

functional / analysis terminal score isolation

Required Avery Red source and fresh analysis were unavailable.

functional / analysis draw and root boundary

Jordan's seeded Draw replay at step 41 was reachable, but visible creation returned "Analysis name is required" while the name input remained in a hidden analysis panel.

functional / analysis position comparison

Required shared A/B/C/D Avery study could not be created.

functional / analysis ownership boundaries

Required fresh Avery studies could not be created.

functional / analysis stale edit and recovery

Required fresh Avery study in two tabs could not be created.

functional / analysis operation receipts

No successful analysis creation/edit request could be captured; visible creation was blocked by the hidden name field.

functional / analysis rejections are atomic

Required real analysis creation/edit setup was unavailable.

functional / analysis source snapshot independence

Required fresh Avery Red win and source study were unavailable.

functional / restart persistence and seed idempotence

R0 could not be prepared because New game and saved-study/report/transplant fixtures were unavailable; R1-R4 were therefore unexecuted.

functional / runtime manifest routes

Manifest correctly documented node startup, SQLite path, and observed auth/bootstrap/archive/Undo/Redo routes, but no New game route or successful analysis/transplant/tactical browser routes were observed; documentation could not be fully matched.

functional / transplant preview rederived positions

Required Red source and analysis fixture were unavailable.

functional / transplant atomic commit mapping

Required shared Transplant preview was unavailable.

functional / transplant recursive merge identity

Required shared Transplant fixture was unavailable.

functional / transplant same study frozen subtree

Required fresh analysis fixture was unavailable.

functional / transplant illegal descendant rollback

Required source and destination studies were unavailable.

functional / transplant dual revision conflicts

No valid Transplant preview or commit request could be captured.

functional / transplant durable receipts

No successful UI Transplant commit was available for receipt testing.

functional / transplant private validation

Required Avery studies and successful preview were unavailable.

functional / tactics immediate and horizon

Required fresh saved studies were unavailable.

functional / tactics forced fork search

Required fresh 2,7,3,7 study was unavailable.

functional / tactics opponent defence and loss

Required fresh tactical studies were unavailable.

functional / tactics full columns and terminal draw

Required fresh study/report setup was unavailable.

functional / tactics complete bounded proof

Required fresh empty-root report was unavailable.

functional / tactics readonly repeatability

Required shared tactical study/report was unavailable.

functional / tactics server input boundaries

No successful Avery tactical request could be captured.

functional / independent same name analyses

The UI could not create even one analysis because the visible creation flow reported a required name while its name field was hidden.

polish / archive replay usability

Gate passed. Jordan's replay showed a separate Yellow-wins view, 0/8 and 42 labelled cells that changed by step, while the active game remained unchanged. Previous, Next, and Close controls were absent.

polish / responsive layout and feedback

At 375px there was no horizontal overflow, but New game and replay navigation were absent. The practice analysis also showed a visible TypeError and no populated analysis board/tree.

polish / reduced motion usability

Reduced motion was emulated and transitions were reduced to about 0.00001s; replay-step text remained understandable. Required populated analysis observations were unavailable after the analysis render error.

polish / control labels and focus

Visible labels and 2–3px focus indicators worked, and Tab/Shift+Tab reached enabled controls. Required Previous/Next/Close controls and populated analysis move/tree controls were missing.

polish / transplant control feedback

The own Jordan study saved only its root. The analysis board was empty, branch selectors had no options, Commit preview was disabled, and the render TypeError prevented preview/commit lifecycle evidence.

polish / tactical proof keyboard inspection

Depth and Generate report controls existed, but Report node had no options and no tactical report, proof branch, or inspected 42-cell board could be opened.

polish / analysis keyboard actions and feedback

The own study had no rendered analysis board or column buttons, no valid child for Undo/Redo, an empty variation tree, and empty comparison selectors, so the required keyboard action sequence could not be completed.

visual / visual typography

Clear type scale and consistent weights distinguish the title, account, totals, turn state, history, analysis tree, comparison, and report. Minor issues include duplicated list numbering and awkward wrapping of long replay metadata on mobile.

visual / visual spacing and layout

Desktop panels, boards, history, tree, comparison, branch preview, and report cards use consistent gutters and alignment. Mobile stacks naturally without horizontal overflow, though long metadata and the extensive analysis flow create some vertical awkwardness.

visual / visual overall craft

The sign-in, game, replay, and populated analysis surfaces share a polished rounded-card and board-game language. However, the normal analysis create/open path exposes a runtime rendering error and can leave required analysis content blank, making the overall experience feel unfinished.

visual / visual responsive consistency

Palette, typography, controls, hierarchy, boards, replay, analysis tree, comparison, branch preview, and tactical report remain coherent at 1280px and 375px, with no horizontal overflow. Long titles, source text, and timestamps wrap less gracefully at mobile size.

Gemini 3.7 Flash

functional / keyboard focus and activation

Keyboard navigation and activation worked, but keyboard Undo/Redo left focus on BODY instead of retaining usable control focus.

functional / analysis rejections are atomic

Most invalid probes were atomic 4xx, but malformed operationId 'nope' was accepted with 200 and mutated the study.

functional / transplant private validation

Root, foreign, unknown, and extra-field probes were 400 and Jordan received 404, but malformed commit operationId was accepted and consumed a preview.

functional / tactics opponent defence and loss

Both required defence/loss positions returned not-established outcomes instead of the specified forced losses.

functional / tactics full columns and terminal draw

Depth-2 recommendations incorrectly included the full column, although terminal draw and win reports were observed.

functional / tactics readonly repeatability

Repeated depth-3 reports were deterministic and read-only, but selecting another node left the old report visible instead of clearing/replacing it.

functional / tactics server input boundaries

Invalid encoded/string-like depth and numeric bounds were rejected, but extra board/account fields were accepted.

polish / responsive layout and feedback

At 375px, the Saved Analyses tab and archive table were clipped. The analysis board/controls and comparison panels expanded to about 523px, placing column 7 and comparison content off-screen.

polish / transplant control feedback

Preview and Commit were clearly shown, but while the first commit response was held the Commit button remained enabled. A second physical activation generated a duplicate commit request; after completion focus fell to BODY instead of a useful tool control.

polish / analysis keyboard actions and feedback

Column keyboard navigation and two drops worked with focus remaining on column 2. Undo/Redo were keyboard-operable and announced, but after Redo became disabled focus fell to BODY and the saved-feedback status element remained hidden/empty rather than providing visible confirmation.

visual / visual spacing and layout

Desktop grouping and gutters are strong, with clear board/history and board/tool columns. Minor unevenness appears in long selected-history areas and narrow analysis/transplant panels.

visual / visual hierarchy and scanability

Primary game state, board, turn/result, totals, replay controls, selected history, comparison result, and tactical proof content are easy to locate; dense proof-tree and narrow tab layouts slightly compete for attention.

visual / visual overall craft

A consistent crafted component language carries through sign-in, game, replay, analysis, comparison, transplant preview, and tactical proof UI. The main blemish is fixed-width analysis/table content that feels less finished on mobile.

visual / visual responsive consistency

Sign-in, active game, and replay adapt well at 375px, but analysis/comparison and transplant surfaces retain roughly 523px-wide boards/panels, clipping columns, tabs, descriptions, and table content horizontally.

Claude Haiku 4.5

constraints / local refresh and health

GET /api/health succeeded with 200 and GET /api/game returned 200 with the bearer token, but a full reload left the page blank and produced a fatal TypeError in app.js, so the authenticated workspace did not render or remain usable.

functional / seeded state import and isolation

Signed-out content was hidden; wrong password returned 401/Invalid email or password; correct sign-in returned a bearer token and protected game read. Avery and Jordan seeded totals/archive data matched, but required seeded pieces rendered at Row 1 instead of Row 6.

functional / completed match archive and replay

Played the shared Red win and observed archive, undo/redo and score changes. Replay exposed no separate replay grid, and page reload produced a blank shell with a null-status TypeError.

functional / gravity turns and full column

Played 4,4,4 and six drops in column 1; full-column feedback worked and reset cleared it. Rendered piece rows were inverted relative to the required coordinates.

functional / horizontal and vertical wins

Shared horizontal and both vertical sequences were executed with histories and score increments. Winning coordinates were row-inverted, and the Yellow sequence visibly retained Red wins! as the status.

functional / both diagonal wins

All four prescribed diagonal sequences were played. Red archives were created, but Yellow attempts showed stale Red wins! status and rendered coordinates did not match required rows.

functional / yellow win and terminal lock

Played 7,1,7,2,6,3,6,1? Actual required 7,1,7,2,6,3,6,4 and terminal clicks were attempted. Archive recorded Yellow wins, but visible status incorrectly remained Red wins!.

functional / exact draw and terminal lock

Played all 42 prescribed drops and an extra drop; Draw, 42 history entries, no markers and terminal locking appeared. Undo left the Draw archive record present instead of removing it.

functional / move history undo redo branching

Executed 4,5,4 branching, two Undo actions, reload attempt, Redo and branch move; operation identifiers were distinct. Turn/reload exactness failed, including the blank-shell reload failure.

functional / new game reset and persistence

Initial and post-win New game resets, column 7, and reload were attempted. Resets cleared board/history and preserved totals, but reload repeatedly produced an empty page with a null-status TypeError.

functional / keyboard focus and activation

ArrowRight, End, Home, ArrowLeft navigation and Enter/Space activation worked for column 2. Rerender moved focus away from the column, violating focus retention.

functional / multi tab revision and duplicate guard

A second tab could not provide the required independently signed-in usable context: it loaded a blank/crashed shell under the shared Playwright context. Duplicate, stale-tab and pending-repeat checkpoints were therefore unverified.

functional / cross tab sign out revocation

The required independently isolated second session could not be established; the second tab blanked under shared storage. Distinct-token revocation and old-token replay were unverified.

functional / accessible grid and winning names

The empty and winning boards exposed 42 grid cells and winning names included color and winning state. Required winning cells were named Row 1 rather than the required Row 6.

functional / analysis fork exact prefix

After opening the completed archive, normal UI exposed only competitive controls and Replay; no labeled analysis creation control or study UI was available.

functional / analysis sibling paths and navigation

No analysis study/tree/navigation UI was exposed, so root children, selection, redo and reuse checkpoints could not be set up.

functional / analysis nested continuations

No analysis continuation/tree UI was exposed; the shared B branch could not be created.

functional / analysis terminal score isolation

No analysis practice UI or study creation control was exposed, so terminal analysis isolation was not testable.

functional / analysis draw and root boundary

No analysis creation/open control was exposed for the seeded Draw archive; root-boundary checkpoints were unverified.

functional / analysis position comparison

No analysis branches or comparison selectors were exposed, so board differences and cursor isolation were unverified.

functional / analysis ownership boundaries

No analysis list, study identity or node UI was exposed; ownership probes could not be grounded in real UI-captured study IDs.

functional / analysis stale edit and recovery

No analysis edit UI or study was available, so the two-tab stale-analysis setup was blocked.

functional / analysis operation receipts

No analysis creation or edit request could be captured because the application exposed no analysis UI route.

functional / analysis rejections are atomic

No supported analysis setup or observed analysis contract was available for the required controlled rejection matrix.

functional / analysis source snapshot independence

No analysis fork/practice UI was exposed, so source snapshot independence could not be exercised.

functional / restart persistence and seed idempotence

Not run: required analysis snapshots, tactical report and uncommitted transplant preview could not be prepared. No lifecycle restart helper was invoked.

functional / runtime manifest routes

Manifest correctly documented node /app/server.js, /app/dropline.db, and observed core routes such as signin, signout, game, archives, move, undo, redo and newgame. Analysis/transplant/tactical routes were not observed in browser requests, so the full route check was unverified.

functional / transplant preview rederived positions

No analysis/transplant UI was exposed; no legitimate source/destination studies or preview could be created.

functional / transplant atomic commit mapping

No transplant preview or commit control/request was available.

functional / transplant recursive merge identity

No transplant UI or study tree was available for recursive merge testing.

functional / transplant same study frozen subtree

No same-study transplant setup or controls were exposed.

functional / transplant illegal descendant rollback

No transplant preview UI or legitimate studies were available for illegal-descendant rollback testing.

functional / transplant dual revision conflicts

No legitimate transplant preview/commit request could be captured for controlled stale-preview probes.

functional / transplant durable receipts

No successful transplant operation receipt could be captured.

functional / transplant private validation

No transplant contract or UI was exposed, so private validation probes lacked legitimate IDs and requests.

functional / tactics immediate and horizon

No tactical report control or analysis study UI was exposed.

functional / tactics forced fork search

No tactical report/proof UI was exposed, so forced-fork setup and proof browsing were unverified.

functional / tactics opponent defence and loss

No tactical study/report UI was available for either defence sequence.

functional / tactics full columns and terminal draw

No tactical report UI was exposed for full-column or terminal-draw checks.

functional / tactics complete bounded proof

No complete tactical proof/report response or visible proof controls were available.

functional / tactics readonly repeatability

No tactical report could be generated or repeated from a saved analysis node.

functional / tactics server input boundaries

No actual tactical request was observed from the UI, so controlled server-boundary probes were unavailable.

functional / independent same name analyses

No analysis creation UI was exposed, so two independently intended same-name studies could not be created.

polish / archive replay usability

Gate passed. Replay cells are unnamed, step changes do not update their visual state, and Previous/Next remain enabled at step 8.

polish / responsive layout and feedback

Game, sign-in, archives, and replay fit at 375px without overflow or overlap, but required analysis surfaces are absent.

polish / reduced motion usability

Reduced motion was honored, but replay board state did not visibly update and analysis surfaces are absent.

polish / control labels and focus

Core controls are named, but the replay range and cells lack accessible names, modal focus is poorly managed, and no visible focus change was observed.

polish / transplant control feedback

Required Jordan analysis/study surface is absent; the workflow could not be run.

polish / tactical proof keyboard inspection

Required Jordan study, tactical report, proof nodes, and explanation branch are absent; the workflow could not be run.

polish / analysis keyboard actions and feedback

No analysis tree, history, or comparison surface is exposed, so the required keyboard checks could not be run.

visual / visual typography

Desktop sign-in, game, history, and replay use readable, consistent type hierarchy for title, account, status, totals, and secondary labels. The required analysis/comparison typography is absent, preventing full credit.

visual / visual color and contrast

Cohesive light palette with clear blue board, white cells, distinct Red/Yellow pieces, dark text, and purposeful status/button accents. Yellow-on-light-gray secondary text is slightly weaker, and analysis surfaces are missing.

visual / visual spacing and layout

Desktop cards, board, controls, totals, history, archives, and replay modal are aligned and comfortably spaced; mobile stacking is orderly. No analysis tree, comparison, transplant, or tactical panels render, leaving required layout coverage incomplete.

visual / visual hierarchy and scanability

Account, game title, status, board, controls, totals, history, archives, and replay are easy to locate, with replay clearly modalized. Selected analysis, sibling tree branches, comparison, and tool outputs are not present.

visual / visual overall craft

Sign-in, game, history, archive, and replay share a consistent simple component language and finished card treatment. The required analysis workspace is absent; analysis creation returned an error and transplant/tactical endpoints reported not implemented. The browser gate itself passed with protected data hidden signed out, wrong password rejected, and bearer-authenticated game state loaded.

visual / visual responsive consistency

Sign-in, game/history, and replay remain visually coherent at 1280×800 and 375×760; headers, scores, history cards, controls, and replay modal adapt naturally. Analysis/comparison and tool panels are absent at both sizes, so full responsive consistency cannot be credited.

Run and source index

Oracle: run-e774bb6c-482c-4c2f-9556-4e7b711b28eb / dropline-four-connect__Zs5Xeyz

GPT-5.4 mini (high): run-419a24fc-0ad9-49f8-9e5e-9f2bb074af37 / dropline-four-connect__ix4VYyy

Gemini 3.7 Flash: run-529333a1-1889-49d5-90fe-edc0b7a5fc4b / dropline-four-connect__JW8NdRB

Claude Haiku 4.5: run-30cde599-faf1-4876-afd2-2d67c76512ab / dropline-four-connect__GXVYjdh

No-op: run-e774bb6c-482c-4c2f-9556-4e7b711b28eb / dropline-four-connect__zndFqyn

Platform task checksum: eb8032690866af12d7cd5f5db6611f63b9b3cad39b5b15f11cb0213a6d014e21

Task ZIP SHA-256: 577451c9d67ef834db1c8366af284b32323ad4f7abc20e396c23c562dd1d63a

All 39 task files match current source; the 12 prompt/judge/runner/reward hashes match each graded export; Oracle solution files match the golden source. Original exports are untouched. Job ZIP copies redact detected provider credentials; score files, verdicts and reasoning are preserved. Full archive hashes and any exclusions/redactions are recorded outside the upload folder in submission-preparation-20260914/package-audit.json.

Fresh final-archive local standard/upload checks are recorded alongside the packaging audit. Historical r6 local evidence includes 49 regression groups, four shared browser journeys and the pending-pointer regression. Packaging did not rerun those browser tests, rebuild Docker images or start paid runs. No claims of new runtime validation are made from historical local checks.